

# Exploratory Data Analysis - Titanic Dataset

## Dataset Overview

The Titanic dataset contains 891 rows and 12 columns. Each row represents a passenger, and columns include details such as:

- PassengerId: Unique ID for each passenger
- Survived: Survival status (0 = No, 1 = Yes)
- Pclass: Ticket class (1 = 1st, 2 = 2nd, 3 = 3rd)
- Name, Sex, Age: Demographic info
- SibSp, Parch: Number of siblings/spouses and parents/children aboard
- Ticket, Fare, Cabin: Travel details
- Embarked: Port of embarkation

Missing values were found in:

- Age: 177 missing values
- Cabin: 687 missing values (most missing)
- Embarked: 2 missing values

## Demographic Analysis

### 1. \*\*Sex Distribution:\*\*

- Males outnumber females significantly.
- Male passengers: ~65% of the dataset
- Female passengers: ~35%

### 2. \*\*Class Distribution:\*\*

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- Most passengers were in 3rd class, followed by 1st and 2nd class.
- Pclass 3: Largest group, often with lower survival rates.

## 3. \*\*Age Distribution:\*\*

- Most passengers were between 20 and 40 years old.
- There were also children and elderly passengers.
- A KDE plot showed a peak around age 30.

## Fare Distribution

Fare data was right-skewed with a few high-value outliers.

- Boxplot showed that some passengers paid significantly higher fares.
- These outliers were mostly 1st-class passengers.
- Most fares were under 100 units.

## Survival Analysis

### 1. \*\*By Sex:\*\*

- Female passengers had a much higher survival rate.
- Most males did not survive.

### 2. \*\*By Class:\*\*

- 1st-class passengers had higher survival rates.
- 3rd-class passengers had the lowest.

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## 3. \*\*Visual Patterns:\*\*

- Count plots with hue='Survived' showed clear differences in survival between sexes and classes.
- A scatter plot of Age vs Fare colored by Survived showed clusters of survivors mostly in higher fares and mid-age groups.

## Correlation Analysis

Using a heatmap of numerical features:

- 'Survived' positively correlated with 'Fare' and negatively with 'Pclass'
- 'Age' had a mild correlation with 'Survived'
- 'SibSp' and 'Parch' showed some relationship, possibly indicating family groups
- These insights can help in building predictive models or further hypothesis testing.